

FROM TRACES TO GUARDED PROGRAMS: EVIDENCE-GATED COMPILATION OF RECURRENT AGENT WORKFLOWS

Anonymous authors

Paper under double-blind review

ABSTRACT

Tool-using agents often invoke a model at every step, even along recurrent read-only paths. Replacing repeated reasoning with deterministic execution could reduce inference cost, but recurrence alone does not make substitution safe: tool arguments may depend on prior observations, apparently read-only operations may hide effects, and replaying the same calls may still alter the final answer. Recurrence identifies an optimization opportunity, not permission to remove reasoning. We introduce *guarded agentic compaction* (GAC), an evidence-gated approach that compiles recurrent execution paths into deterministic programs only when substitution is justified. GAC compiles initial read-only prefixes only: it grounds tool arguments in observable state, treats writes, approvals, handoffs, and unknown effects as hard barriers, protects compiled execution with input, output, and environment guards, and admits specialization only when finite-sample evidence satisfies configured risk and confidence requirements. Otherwise, it retains the largest justified prefix when possible or falls back to the unchanged agent. Those admission certificates are per candidate; extending them compiler-wide requires freezing the candidate search.

Across three live GitHub workflow families and 90 unseen test cases, GAC satisfies all 90 task requirements, compared with 89/90 for the unchanged agent, while reducing model requests by 66.6%, tokens by 63.1%, latency by 64.2%, and estimated cost by 58.7%. A time-forward evaluation across five frozen public repositories specializes four and refuses one. On four public trace benchmarks, NESTFUL, API-Bank, and BFCL v4 contain recurrent, replayable traces but insufficient qualifying evidence to satisfy GAC’s default 0.05 risk bound at the required confidence, whereas AppWorld provides sufficient evidence for admission. Even after admission, GAC abstains from dispatch when runtime behavior changes the execution structure or violates its entry requirements, falling back to the unchanged agent.

GAC reframes agent optimization as evidence-gated specialization: compile only what is justified, dispatch only when runtime conditions remain valid, and preserve model reasoning everywhere else.

1 INTRODUCTION

The standard agent loop pays for a model decision at every tool boundary (Yao et al., 2023). That flexibility is valuable while a workflow is novel, but mature agents often revisit the same read-only evidence path with the same data dependencies, where repeating the decision adds latency, tokens, and cost without adding adaptation.

Removing that decision is not ordinary caching. A tool argument may depend on a specific earlier observation, a repeated sequence may cross an approval or write, an apparently read-only tool may hide an undeclared effect, and replaying the same outputs may still change the downstream answer. The problem is not “find recurring traces” but deciding when a recurrent model-mediated region may be replaced by a deterministic program, and when it must be refused.

Consider held-out issue #4420. Given only `issue_number=4420`, the observed trace calls `record(4420)`, passes the *returned* `record.issue_number` to `labels` and `comments`, and invokes the latter with `limit=100`. All 132 discovery executions take that route, so a recurrence-only optimizer would emit all three reads. GAC instead retires the *region*: `comments.limit` is reachable

Two cohorts, two substitution failures

Expanded issue #4420 — multi-node metric evaluation
 record → labels → comments(limit=100)

Earlier artifact: #6602

recur + replay → ship
 45/45 tool replays
 4 requests → 1
 tool contract passes

Answer miss: #6602

expected: Markdown link
 compiled: link text only
tool contract = pass
answer contract = fail

Expanded compiler: stop where provenance ends

full candidate: RETIRE
 comments.limit: ungroundable
 no consistent expression
 emit record → labels
 comments + render stay with agent
92/92 · U=.0498 ≤ α=.05

Earlier continuation check

detects evidence mismatch
 render check: 18/18 pass
 provider-free mechanism check;
 not live safety evidence

Figure 1: Two retained cohorts expose different failure modes: issue #4420 forces a shorter grounded prefix, and issue #6602 shows that clean tool replay need not preserve the downstream answer.

neither from the entry state nor from either earlier result, so one unwitnessed slot blocks the three-read candidate as `ungroundable_slot`. What survives is the maximal justified prefix `record → labels`; the agent still chooses the limit and writes the answer. That two-read artifact passes 92/92 calibration groups at $\alpha = 0.05$ (upper bound 0.0498, a per-candidate certificate).

Why stop there? An earlier 18-record study supplies the counterexample: its three-read artifact replayed all 45 tool calls, yet issue #6602 lost the URL from a Markdown link and only 17/18 downstream answers stayed exact. A provider-free continuation guard detects and checked-renders that miss — a mechanism check, not live safety evidence. Together the two cohorts (fig. 1) show why recurrence only proposes a candidate.

The closest precedent is the tracing just-in-time compiler: Dynamo (Bala et al., 2000) detects a hot path, compiles it behind a *guard*, and deoptimizes when the guard fails (Hölzle et al., 1992). Agent traces admit the same structure — hot-trace detection becomes family mining, guard synthesis an entry contract, deoptimization fallback to the unchanged agent — but add three obligations, and claim one capability less: a JIT enters a trace at an arbitrary program point, whereas GAC compiles only *initial* read-only prefixes (eq. (2)), because its runtime resolves artifacts at the entry boundary. The obligations are that tool arguments be shown *grounded* in observable state rather than invented; that declared *effects* permit speculation, since an agent’s “interpreter” can call the outside world; and that the residual error rate, being statistical, needs a finite-sample bound rather than a soundness proof.

We study these obligations through *guarded agentic compaction* (GAC), a compile-or-retire trace-to-program system for recurrent read-only prefixes (fig. 2). The core contribution is not “agent compilation” in general, but an admissibility argument for trace-derived specialization:

1. A guarded specialization pipeline (Alg. 1) combining typed provenance, effect and position barriers, bounded synthesis, runtime verification, and finite-sample compile-or-retire admission (Alg. 2, proposition 1); every stage can refuse, and the default output is the unchanged agent.
2. A real-provider evaluation across three public-record workflow families, plus a narrower cross-repository, time-forward extension keeping both artifacts and principled retirements.
3. A negative-result boundary: recurrence and replayability do not license deployment. NESTFUL, API-Bank, and executed BFCL gold plans retire under the same admission logic, and hand-written programs are competitive or tied wherever the workflow is already known.

4. A pre-registered fourth substrate, AppWorld — the first external corpus whose sample size makes admission reachable, and which admits — plus a measurement on 8,190 released baseline trajectories where the admitted region is eligible on 2,339/2,340 full-code but only 1/4,680 ReAct or plan-and-execute trajectories, separating *admissible* from *dispatchable*.

2 PROBLEM FORMULATION

Episodes and candidate regions. An episode $E = (z, M, e_{1:T}, y)$ contains an entry-state snapshot z , a content-addressed execution manifest M , ordered events e_t (model requests and responses, tool calls and results, handoffs, guardrails, approvals, errors, commit boundaries), and an observable outcome y . Each tool f carries an application-owned effect declaration $\epsilon(f)$ and capability flags such as *speculatable*; an unknown effect is not a read.

A candidate region $R = [a, b]$ begins at a model-request boundary and ends after a tool result. It is *groundable* when every call argument is either an application-declared literal for that slot or an expression over entry state or prior results inside R ,

$$\forall c_j \in R, \forall u \in \text{args}(c_j) : \text{Lit}(c_j, u) \vee \exists s \in \text{Src}(z, o_{<j}), \exists g_{j,u} \in \mathcal{L} : u = g_{j,u}(s), \quad (1)$$

where Lit is a schema-level allowlist fixed independently of the traces, $\mathcal{L}$ is a closed, bounded transform library, and $\text{Src}(z, o_{<j})$ enumerates the flattened entry-state paths and each prior result $o_i, i < j$; and *effect-admissible* when every tool in R is declared read-only, speculatable, replayable, and inside one principal and isolation partition. The witness is deliberately *single-source* — one chain over one observation, not a join over the history — which is the quantifier Alg. 3 implements. Handoffs, approvals, writes, errors, and unknown effects are hard barriers. Because the deployed runtime resolves artifacts at the initial model boundary, the compiler also imposes a *position invariant*:

$$a = 0 \quad \text{in the normalized model-boundary index.} \quad (2)$$

This is not aesthetic. A suffix program dispatched at entry can reorder or duplicate calls the agent has already made — a semantic mismatch between compilation-time and resolution-time position rather than a tuning failure, and a pilot that violated eq. (2) produced exactly that fault (appendix E). Everything that follows is therefore an *initial read-only prefix* compiler, not the arbitrary-entry trace compiler a JIT is; section 5.3 prices what that costs.

Selective optimization objective. The compiler emits an artifact $A = (P, H, V, q, \eta, M)$: a deterministic program, hard guard, output verifier, nonconformity score, threshold, and manifest pins. Let $x = (z, M', \omega)$ be a future entry state, runtime manifest, and external state, where $M' \simeq M$ is equality on every pinned compatibility field and ω affects outcomes and cost but is unobservable at the dispatch boundary. Dispatch is *selective*,

$$d_A(x) = \mathbf{1}\{H(z, M') = 1 \wedge q(z) \leq \eta \wedge M' \simeq M\}, \quad (3)$$

and any failed precondition runs the unchanged baseline B . If execution fails before a commit and staging proves the attempt clean, B runs; a dirty failure after a commit is an incident, not a fallback. Fix $L(A, x) = 1$ for an externally visible task/contract violation or incident after dispatch, $L = 0$ for a clean abort followed by unchanged B , and let $C(A, x)$ be the complete realized cost of the attempt including a clean fallback. For a future group $G = (x_1, \dots, x_m)$, set $D_A(G) = \max_i d_A(x_i)$ and $W_A(G) = \max_i d_A(x_i)L(A, x_i)$. We seek

$$\max_A \mathbb{E}_{G \sim \mathcal{D}_{\text{grp}}} \left[\sum_{x \in G} d_A(x) \{C(B, x) - C(A, x)\} \right] \quad \text{s.t.} \quad \Pr_{G \sim \mathcal{D}_{\text{grp}}} [W_A(G) = 1 \mid D_A(G) = 1] \leq \alpha. \quad (4)$$

Both halves take the same group measure $\mathcal{D}_{\text{grp}}$, the objective summing within a group rather than over pooled episodes, so that unequal groups are not weighted differently by objective and constraint. Three properties shape everything that follows. *First*, the constraint conditions on dispatch: abstention does not enter its denominator, although it earns no savings in the objective; by convention its conditional risk is zero when population dispatch probability is zero. *Second*, returning *no* artifact is feasible, so *RETIRE* is a legitimate output, and eq. (4) is a design target rather than a solved program — the implementation calibrates a bounded ranked set and retains nondominated survivors, and we prove neither optimality nor a compiler-wide guarantee for that search. *Third*, L scores the *episode*: a sound prefix whose baseline continuation errs still contributes $L = 1$, so the gate certifies end-to-end compliance under substitution, not non-inferiority, and admission requires B to already succeed on a $1 - \alpha$ share of the dispatched slice (appendix G).

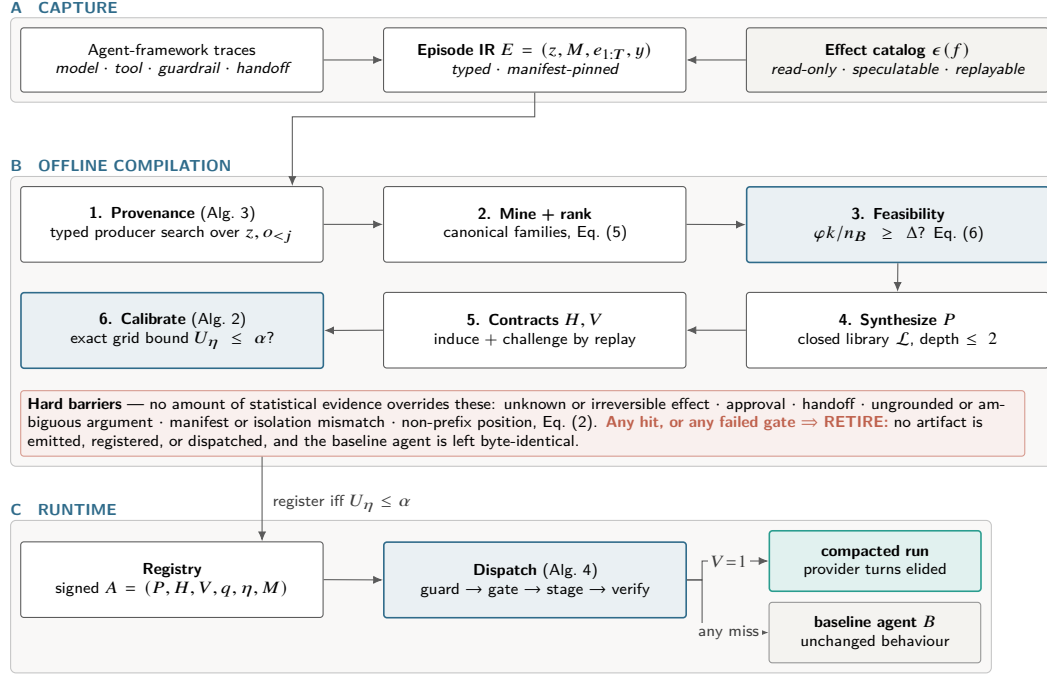


Figure 2: Architecture of GAC. (A) Capture normalizes traces into one typed Episode IR against a declared effect catalog. (B) Compilation is a cascade of independent rejection points; the shaded band lists barriers no statistical evidence can override. (C) At runtime one terminal edge compacts; the rest return the unchanged agent.

Algorithm 1 COMPILER — the compile-or-retire cascade. Every candidate is either registered or assigned an attributed RETIRE, so refusal accounting remains explicit. Stages marked (CALLER) are separate library entry points run in this order rather than steps inside one function.

Input: episodes $\mathcal{E}$; effect catalog C ; entry schema Θ ; transform library $\mathcal{L}$; ambiguity cap κ ; mining window $w_{\max}$; budgets α, δ ; grid Λ ; feasibility target Δ

Output: artifact set $\mathcal{A}$, or RETIRE with attributed reasons

```

1:  $\mathcal{A} \leftarrow \emptyset$ 
2:  $\mathcal{E} \leftarrow \text{QUALIFY}(\mathcal{E}, C)$ ;  $\{\mathcal{E}_M\} \leftarrow \text{PARTITIONBY}(\mathcal{E}, \text{manifest}, \text{principal}, \text{isolation})$  ▷ drop partial runs, unpinned manifests
3: for all partitions  $\mathcal{E}_M$  do
4:    $(\mathcal{T}, \mathcal{D}, \mathcal{K}, S) \leftarrow \text{GROUPEDSPLIT}(\mathcal{E}_M)$  (CALLER) ▷ train / dev / calibration / sealed test, split by group
5:    $G \leftarrow \text{BUILDPATG}(\mathcal{T}, \Theta, C, \mathcal{L}, \kappa)$  ▷ Alg. 3: block ungrounded or ambiguous slots
6:    $\mathcal{F} \leftarrow \text{CANONICALIZE}(\text{MINEREGIONS}(G, C, w_{\max}))$ , keeping only cross-group support
7:   if  $\text{CEILING}(\mathcal{T}, \mathcal{F}) < \Delta$  then (CALLER)
8:     record RETIRE (infeasible ceiling, Eq. (6)); continue ▷ try the next compatible partition
9:   end if
10:  for all families  $F \in \mathcal{F}$  ranked by Eq. (5) do
11:     $P \leftarrow \text{SYNTHESIZE}(F, \mathcal{L})$  at depth  $\leq 2$ ; if  $P = \perp$  then record RETIRE (synthesis); continue
12:     $(H, V) \leftarrow \text{INDUCECONTRACT}(F)$ ; if  $\neg \text{REPLAYANDCHALLENGE}(P, V, \mathcal{D})$  then record RETIRE (counterexample); continue
13:     $q \leftarrow \text{FITSCORE}(\mathcal{D})$ ; freeze  $q$  ▷ dev groups only; never calibration
14:     $\eta \leftarrow \text{CALIBRATE}(P, H, V, q, M, \mathcal{K}, \Lambda, \alpha, \delta)$ ; if  $\eta = \perp$  then record RETIRE (risk); continue ▷ Alg. 2
15:     $\mathcal{A} \leftarrow \mathcal{A} \cup \{\text{PACKAGE}(P, H, V, q, \eta, M)\}$  with evidence and lifecycle
16:  end for
17: end for
18: return  $\text{SELECTUNDOMINATED}(\mathcal{A})$  if nonempty, else RETIRE

```

3 GUARDED AGENTIC COMPACTION

Figure 2 and Alg. 1 give the same six stages at two levels of detail. The order is part of the argument: each stage can only weaken a candidate, so a later stage may reject one an earlier stage passed, but never authorize one an earlier stage blocked.

1. Typed provenance. A capture adapter normalizes framework traces into the Episode IR; manifests pin prompt, policy, guardrail, tool, model, SDK, tracer, entry-contract, and effect-catalog

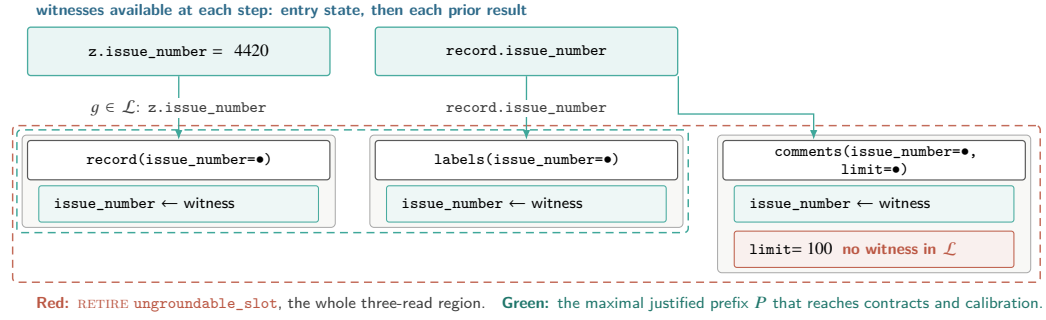


Figure 3: Equation (1) on the held-out issue-4420 record. Every argument slot needs an expression in $\mathcal{L}$ over the entry state or a prior result; the two `issue_number` slots have one and the `limit` slot has none. The question is strictly harder than recurrence — not “did this sequence repeat?”, which all 132 discovery executions did, but “can every socket be wired from an allowed prior value?” One unwitnessed slot retires the whole region, and so does a slot with more than κ witnesses.

identities, and episodes with different compatibility keys are partitioned before any learning. For each call-argument slot, the program-argument trace graph (PATG, Alg. 3) searches entry-state and prior-result paths for typed values reachable by a bounded transform — execution provenance in the database sense (Cheney et al., 2009), specialized to tool arguments. Two outcomes block a region rather than guessing: a slot with *no* witness is a genuine model decision, and one with more than $\kappa = 3$ witnesses is ambiguous (fig. 3). A repeated tool name, a matching literal, and a successful replay are not witnesses; only an expression in $\mathcal{L}$ is.

2. Region mining and ranking. The miner enumerates model-boundary-to-result windows of at most $w_{\max}$ tool calls, qualifies live-ins and effects against a human-signed rather than inferred catalog, and hashes tool signatures, dependency topology, run shape, and live-in/live-out shape while abstracting literals. Families require support across independent scenario *groups*, not raw event frequency, so one chatty scenario cannot manufacture recurrence. Survivors are ranked by an MDL-inspired trade-off that pays for expected savings and charges for heterogeneity, effect exposure, and program size:

$$\text{score}(F) = \underbrace{s_F \bar{k}_F c_m}_{\text{support-weighted saving}} - \lambda_1 \text{Ent}(F) - \lambda_2 \rho_{\text{eff}}(F) - \lambda_3 |F|, \quad (5)$$

with s_F supporting train groups, $\bar{k}_F$ the mean removable model requests, c_m their unit cost, and $\text{Ent}(F)$ the Shannon entropy over canonical variants inside the family; without λ_1 the ranking prefers a heterogeneous family whose canonical form fits none of its members. Equation (5) affects only *which* family is tried first.

3. Feasibility ceiling. Before any synthesis runs, an estimator bounds what the compiler could achieve with a perfect verifier and a gate that never abstains:

$$\Delta_{\max} = \frac{\varphi k}{n_B} \geq \Delta \quad (\text{pre-synthesis filter}), \quad \varphi \rho k \geq \Delta n_B \quad (\text{realized, post-verification}), \quad (6)$$

with n_B baseline model requests per episode, φ the fraction of episodes holding an eligible region, k the requests removed per successful dispatch, and ρ the verifier pass rate. Stage 3 runs before synthesis, so no verifier exists and it evaluates only the left form, which fixes $\rho = 1$; the right form is what the workload must meet once V is induced. Since $\rho \leq 1$, no measured reduction exceeds $\Delta_{\max}$, so a workload under target is declined before a compiler is built for it.

4–5. Bounded synthesis and contracts. Synthesis searches the 23-operator library of table 3 at depth at most two; bindings are ranked by cross-group stability, and a group-wise refit rejects bindings exploiting row-level coincidence. The hypothesis space is deliberately small enough to read — the approach resembles programming-by-example (Gulwani, 2011) and bounded sketching (Solar-Lezama et al., 2006) — and it does *not* generate arbitrary Python. Hard guards then constrain entry

Algorithm 2 CALIBRATE — fixed-grid exact selective admission. The score q and the grid Λ are frozen *before* this procedure sees a calibration group, which is what makes the union bound legitimate.

Input: frozen candidate $A_0 = (P, H, V, q, M)$ without threshold; calibration groups $\mathcal{K}$ with fixed per-instance violations $L(A_0, x)$; grid Λ ; budgets α, δ
Output: threshold η with a simultaneous guarantee, or $\perp$ (RETIRE)
1: $\Pi \leftarrow \emptyset$ ▷ admissible (coverage, threshold) pairs, distinct from Alg. 1’s $\mathcal{A}$
2: **for all** $\eta \in \Lambda$ **do**
3: define $d^\eta(x)$ by Eq. (3) with threshold η ; $D^\eta(g) \leftarrow \max_{x \in g} d^\eta(x)$; $W^\eta(g) \leftarrow \max_{x \in g} d^\eta(x) L(A_0, x)$ ▷ vacuous bound $U_\eta = 1$
4: $K_\eta \leftarrow \{g \in \mathcal{K} : D^\eta(g) = 1\}$; $n_\eta \leftarrow |K_\eta|$; **if** $n_\eta = 0$ **then continue**
5: $k_\eta \leftarrow |\{g \in K_\eta : W^\eta(g) = 1\}|$ ▷ wrong after dispatch only, never abstention
6: $U_\eta \leftarrow 1$ if $k_\eta = n_\eta$, else $\text{Beta}^{-1}(1 - \frac{\delta}{|\Lambda|}; k_\eta + 1, n_\eta - k_\eta)$; **if** $U_\eta \leq \alpha$ **then** $\Pi \leftarrow \Pi \cup \{(\eta, |\mathcal{K}|, \eta)\}$ ▷ Eq. (7)
7: **end for**
8: **return** $\perp$ if $\Pi = \emptyset$, else the η component of $\text{lexmax}_{(\text{cov}, \eta) \in \Pi} (\text{cov}, \eta)$ ▷ largest empirical group coverage; larger threshold breaks ties

types, required fields, categorical sets, numeric intervals, patterns, isolation keys, allowed effects, and manifest pins; verifiers constrain live-out presence, type, cardinality, empirical hulls, provenance, effect multisets, and call counts. These are candidate invariants, not proofs — which is why a statistical gate follows rather than replaces them.

6. Exact risk-gated admission. Algorithm 2 decides deployment. The score q is a logistic model over entry-observable features — missing fields, hull margin, temporal drift, and provenance ambiguity — fitted by regularized maximum likelihood on *dev* groups only, never on calibration groups, then frozen together with the finite threshold grid $\Lambda = \{0.02, 0.05, 0.08, 0.11, 0.14, 0.17, 0.20, 0.25, 0.30, 0.40, 0.50\}$. Two data signals do different jobs. q is fitted to predict an *unproductive* dev group — guard rejection, abstention, or manifest mismatch — never a wrong execution, since REPLAYANDCHALLENGE already retired those; the bound below is counted only on dispatched groups that turned out wrong (appendix C). For each $\eta \in \Lambda$ the compiler counts admitted calibration groups (n_η) and those containing at least one post-dispatch violation (k_η), and inverts the one-sided exact binomial test (Clopper & Pearson, 1934) at a Bonferroni-corrected level, with explicit empty and all-violation cases:

$$U_\eta = \begin{cases} 1, & n_\eta = 0 \text{ or } k_\eta = n_\eta, \\ \text{Beta}^{-1}\left(1 - \frac{\delta}{|\Lambda|}; k_\eta + 1, n_\eta - k_\eta\right), & 0 \leq k_\eta < n_\eta, \end{cases} \quad U_\eta|_{k_\eta=0} = 1 - \left(\frac{\delta}{|\Lambda|}\right)^{1/n_\eta}. \quad (7)$$

It admits the largest-coverage threshold with $U_\eta \leq \alpha$ and retires otherwise — a fixed-grid instance of learn-then-test (Angelopoulos et al., 2022). The closed form on the right is the operative constraint here: at $\alpha = 0.05$, $\delta = 0.1$, $|\Lambda| = 11$, a zero-violation family still needs $n_\eta \geq 92$ admitted groups, and the default output is refusal.

That floor also bounds what selectivity is *certifiable*: coverage $c = n_\eta/|\mathcal{K}|$ needs $|\mathcal{K}| \geq \lceil 92/c \rceil$, so on the $|\mathcal{K}| = 92$ cohorts used throughout — the smallest satisfiable pool — the only admissible coverages are $c \in \{0, 1\}$ whatever q ranks, since abstaining on one group leaves $U_\eta = 0.0503 > \alpha$. Intermediate coverage can still be *observed*; it cannot be certified (appendix C).

Proposition 1 (Per-candidate selective-risk admission). *Condition on train/development data and fix one candidate (P, H, V, q) , group-level violation rule, and finite grid Λ before observing calibration. If calibration groups are i.i.d. from the future-group distribution, then with probability at least $1 - \delta$, every $\eta \in \Lambda$ satisfies $r_\eta \leq U_\eta$, where r_η is $\Pr(W^\eta = 1 \mid D^\eta = 1)$ when population admission is positive and is defined as zero otherwise. Consequently the selected $\hat{\eta}$ satisfies $r_{\hat{\eta}} \leq \alpha$ whenever $U_{\hat{\eta}} \leq \alpha$.*

Appendix A gives the proof. The guarantee is deliberately narrow: *per fixed candidate*, not compiler-wide over the adaptive search Alg. 1 performs, and conditional on group exchangeability rather than establishing it (section 7). Guarded deployment requires an explicit evidence boundary, not just a recurrent trace and a successful replay.

4 EXPERIMENTAL SETUP

The evaluation asks three questions: does specialization preserve outcomes while saving work on real records (section 5.1)? Does it transfer under a time-forward split (section 5.2)? And does it refuse when evidence is thin (section 5.3)?

Table 1: Primary live-provider results ($n = 30$ held-out records per family). “Interfaces” counts provider-visible tool interfaces. All conditions see identical records, model, and grader; *Manual* is hand-written with full knowledge of the workflow, and table 4 prices it on these same resource axes.

Family	Exact held-out contract			Reduction of compiled vs. baseline (%)				
	Base	Compiled	Manual	Requests	Interfaces	Tokens	Latency	Cost
Issue-type routing	30/30	30/30	30/30	50.0	0.0	39.5	51.7	32.0
PR-outcome audit	30/30	30/30	30/30	75.0	66.7	80.8	73.0	75.3
Backlog-attention routing	29/30	30/30	30/30	74.8	66.3	81.4	68.9	75.1
Weighted total ($n = 90$)	89/90	90/90	90/90	66.6	44.2	63.1	64.2	58.7

4.1 BENCHMARK AT A GLANCE

Every GitHub sample begins with one issue or pull-request number from a pinned public snapshot, and the agent must make read-only calls and return exact, source-grounded fields rather than a preference. The three primary families use 132 discovery and 30 balanced held-out records each, with live provider calls, and each pairs a read-only evidence path with an exact answer contract — issue type, pull-request outcome, or backlog attention route, each returning a category plus title and a verbatim excerpt — so the benchmark tests both the structured answer and whether its evidence survives compaction (table 5, appendix E).

Cross-repository time-forward extension. A narrower two-read PR-outcome task spans five frozen public repositories. Held-out records are strictly newer than discovery records, and selection is sealed before provider calls; repositories without an admitted artifact remain in the result. A balanced rerun on three repositories controls the open/merged/closed_unmerged mix.

External compiler substrates. NESTFUL, API-Bank, and executed BFCL v4 gold plans supply executable traces with the argument-level producer structure provenance, synthesis, replay, and admission need, and measure none of them as planning quality (Basu et al., 2025; Li et al., 2023; Patil et al., 2025). AppWorld adds a fourth and different one (Trivedi et al., 2024): its public gold solutions, executed on its pinned resettable backend, give 136 byte-exactly reproducible traces over 68 typed APIs and enough groups for eq. (7) to be satisfiable, and its released baseline runs supply held-out trajectories for the dispatch measurement. No model runs on any substrate, so none licenses an accuracy or efficiency claim and none is pooled with the GitHub families (appendix F.3).

Provider configuration. All live studies use one provider and model — OpenAI gpt-5.6-1una via the openai-agents Responses interface at reasoning effort low — and price tokens from published list rates rather than an invoice. Appendix C gives the full settings and the rate table; conditions within a family differ in structure alone.

Conditions, metrics, and statistics. Each GitHub family compares the unchanged baseline B , the compiled artifact, and a hand-written pre-model program on the same records, so comparisons are paired rather than across cohorts. Quality is exact task-contract satisfaction; resources are requests, interfaces, tokens, latency, and estimated cost. We use paired exact McNemar tests, bootstrap intervals, and Wilcoxon signed-rank tests, and the only guaranteed multiplicity control is the Bonferroni term in eq. (7). All admission runs use $\alpha = 0.05$, $\delta = 0.1$, and $|\Lambda| = 11$ (appendix C).

5 RESULTS

5.1 PRIMARY WORKFLOWS: PRESERVED CONTRACTS, LESS WORK

Across the 90 held-out records, GAC satisfies 90/90 exact contracts versus 89/90 for the unchanged agent (table 1), reducing provider requests by 66.6%, tokens by 63.1%, latency by 64.2%, and cost by 58.7%. Issue-type routing admits only a two-read prefix; the two deeper workflows approach 75% fewer requests.

Table 2: Where refusal happens, and what happens where it does not. Stage numbers are Alg. 1’s; “slots” is the share of dependency slots with a recovered producer, and $n_{\max}$ is the largest family support against the 92 eq. (7) requires. “p/a/w” is pass/abstain/wrong on each substrate’s *family-level* replay split, which is held out from a different fit than the admission split quoted in the text (appendix F.3).

Substrate	Traces	Slots (1)	Windows (2)	At floor (2)	Synth. (4)	Replay p/a/w (5)	$n_{\max}$ (6)	Adm.
NESTFUL	1,415	96.3%	1,207	32	12	24 / 12 / 0	26	0
API-Bank	212	63.5%	37	8	2	0 / 2 / 0	8	0
BFCL v4	200	68.0%	86	9	4	3 / 3 / 0	15	0
AppWorld	136	63.2%	136	33	1	34 / 0 / 0	136	1

No held-out record is a compiled-only failure; the one-sided 95% upper bound on that discordance is 3.3%, which supports preservation on the observed cohort, not equivalence in general. Each artifact was admitted with 92 zero-violation calibration groups ($U_{\hat{\eta}} = 0.0498 \leq 0.05$). Each family mined two candidates and kept the survivor by empirical dominance without freezing the search, so these are *per-candidate* certificates under proposition 1, not compiler-wide ones (corollary 2, appendix C).

Hand-written programs also reach 90/90. Table 4 prices them on the same axes: they tie on requests, issue fewer interfaces (66.5% vs. 44.2%), are marginally cheaper in tokens and dollars, and lose only on latency. We claim discovery and admission, not runtime dominance over code written by someone who already knows the workflow.

5.2 TIME-FORWARD TRANSFER: COMPILE OR RETIRE

On a narrower two-read pull-request outcome task across frozen public repositories, with held-out records strictly newer than discovery ones and selection sealed before any provider call, GAC completes 120/120 exact pairs on four of five repositories and retires `pytorch/pytorch` rather than loosening its gate; requests fall 44.4% and tokens 52.4%. That retirement is the gate’s, not provenance’s: all 116 windows cleared with nothing blocked, but the unenforced discovery pool held 2 merged records against a balanced 10/10/10 test split, and no threshold attained $U_{\eta} \leq \alpha$ (appendix F). A balanced three-repository rerun, which controls that mix, reaches 180/180 exact pairs at 66.7% fewer requests and 78.4% fewer tokens (table 6). A fixed two-read template is more efficient in the five-repository core and essentially tied in the rerun: evidence of discovery and principled retirement, not of runtime dominance.

5.3 WHAT BINDS: SAMPLE SIZE, THEN DEPLOYABILITY

The external substrates isolate the binding constraint (table 2). All four clear provenance, synthesis, and replay with *no* wrong execution, then NESTFUL, API-Bank, and executed BFCL gold plans retire — each settled arithmetically before the compiler ran, which is why a fourth was needed. AppWorld’s gold solutions reach $n_{\max} = 136$, making it the first external corpus where eq. (7) is satisfiable, and it admits one artifact at 92/92 zero-violation groups ($U_{\hat{\eta}} = 0.0498$) with 11/11 exact replay on its sealed admission split, under both pre-registered effect arms. Two features bound it: the gate is step-like, and the admitted program is argument-free, so provenance fixed its *length* rather than its content (appendix F.3).

Admissibility is not dispatchability. Gold solutions are not agent trajectories, so we also test the admitted region against AppWorld’s 28 released baseline runs. Over 8,190 trajectories it is structurally eligible at the entry boundary in 37.9%, and the architecture explains the spread: 2,339/2,340 for full code with reflection and 762/1,170 for iterative parallel function calling, against 1 of 4,680 for ReAct and plan-and-execute, which open by reading API documentation so the prefix leaves position 0 (table 7). These repeat tasks across the 28 runs, so they are descriptive units, not independent samples; a miss abstains cleanly, and the measurement bounds φ from above rather than estimating a dispatch rate.

6 RELATED WORK

Guarded trace and agent workflow compilation. GAC adapts the tracing-JIT pattern—compile a hot path behind a guard, deoptimize on a miss (Bala et al., 2000; Hölzle et al., 1992)—to agents, adding grounded arguments and declared effects (Lucassen & Gifford, 1988). Agent systems mine workflows, cache plans, compile orchestration, validate plans, or prune topology (Abuzakuk et al., 2026; Wang et al., 2025; Zhang et al., 2025; Wei et al., 2026; Winston et al., 2026; Li et al., 2026; Chen et al., 2026); DSPy optimizes prompts, GEPA evolves them, RouteLLM changes models, LLM-Compiler parallelizes calls (Khatab et al., 2024; Agrawal et al., 2026; Ong et al., 2025; Kim et al., 2024). Unlike these, GAC treats recurrence only as a proposal: provenance, effects, position, contracts, and finite-sample risk each retain a veto.

Synthesis, risk control, and evaluation. Bounded search draws on programming by example, sketching, and provenance (Gulwani, 2011; Solar-Lezama et al., 2006; Cheney et al., 2009); admission combines Clopper–Pearson with fixed-grid learn-then-test (Clopper & Pearson, 1934; Angelopoulos et al., 2022). BFCL, ToolLLM, and AppWorld score tool use or interactive agents broadly (Patil et al., 2025; Qin et al., 2023; Trivedi et al., 2024). NESTFUL and API-Bank retain observed intermediate results directly; BFCL and AppWorld become compiler substrates only once gold artifacts run on their pinned backends, and no provider-free row measures agent quality.

7 DISCUSSION AND LIMITATIONS

The supported claim is narrow: repeated read-only prefixes can become guarded programs that preserve held-out contracts on the tested GitHub families and retire when evidence is inadequate. Appendix G expands each limitation.

Gate maturity is the main scientific risk. Proposition 1 is per-candidate, not compiler-wide over Alg. 1’s adaptive search unless that search is frozen (corollary 2); the primary families do not freeze, so a multiplicity repair would raise their requirement to 106 groups (appendix C). The gate is also step-like at $n_\eta \geq 92$, for two reasons this design cannot separate: every cohort fixes $|\mathcal{K}| = 92$, so no intermediate coverage is *certifiable* whatever q ranks (section 3), and q *refuses* correctly far better than it *discriminates*. A pre-registered protocol (appendix H) has since run across five repositories and 240 held-out triples, where a support-only comparator sharing every mechanism but the risk budget is indistinguishable from the learned gate, neither showing graded coverage. The follow-up is to buy calibration groups first: below $\lceil 92/c \rceil$ of them, no refit of q can be read.

Group independence is assumed, not established. Proposition 1 needs i.i.d. groups and neither corpus establishes exchangeability: GitHub groups are single records from one snapshot with `min_days=min_principals=1`, and AppWorld’s task groups share a scenario’s supervisor and database. A seeded simulation (appendix A) prices the exposure — miscoverage inflates well past nominal under block-correlated groups — and the cluster-aware repairs all cost effective sample size.

External validity, dispatchability, and comparison. The richer evidence comes from one revision-pinned repository and one provider/model family, and admission is not dispatchability: eq. (2) is the operative limit, which section 5.3 prices. We show no superiority under drift or across providers, do not price manual authoring, and run no head-to-head against the workflow-reuse line, whose prompt-level induction, parallel scheduling, and plan caching make no admissibility decision to compare against — leaving the *unguarded* ablation unrun. Exact contracts also do not certify a downstream continuation, and writes, handoffs, hosted tools, undeclared effects, and removed-turn guardrails lie outside proposition 1.

8 CONCLUSION

GAC treats recurrence as candidate evidence, not permission to rewrite an agent: it preserves exact held-out contracts while saving resources, retires all families on three external substrates, and admits a small, architecture-limited region on the fourth. Judge workflow optimizers by their evidence, refusals, and fallback, not only by the turns they remove.

AI USE STATEMENT

The authors used generative AI tools to restructure and copy-edit the manuscript, suggest section organization, draft and revise paper text, critique methodological exposition and internal consistency, convert repository-grounded results into LaTeX tables, and assist in designing and implementing the experiment and analysis scripts used to produce the reported results. AI tools were not used to fabricate, hallucinate, or substitute for results: every study was executed against real systems (live GitHub repositories, recorded agent traces, and public benchmark datasets), and every reported number, citation, equation, and comparison was independently checked against the retained execution artifacts in the repository. The authors take full responsibility for the final content, the experimental design, and the validity of all reported results.

ETHICS STATEMENT

The main risk in removing model-mediated decisions is over-trusting a compiled prefix, and thereby accelerating unsafe automation or deleting a guardrail evaluation that happened to run at the removed boundary. The method mitigates this by compiling only read-only prefixes, treating unknown effects, approvals, handoffs, and writes as hard barriers, and defaulting to retirement or fallback when evidence is insufficient. All study data are public records (GitHub repository metadata) or public benchmarks; no personal or private data were collected. The artifact is not a license to bypass human review, legal or compliance controls, or safety guardrails; unsupported regimes are part of the result, not exceptions to hide.

REPRODUCIBILITY STATEMENT

Algorithms 1 and 2 give the compiler and admission protocols, Algs. 3 and 4 give provenance construction and runtime dispatch, appendix A proves proposition 1, and appendix C records the registered statistical and synthesis settings. The accompanying artifact contains source manifests, raw result files, deterministic table and figure generation, and scripts for the live GitHub studies, the cross-repository extension, and all four compiler substrates: NESTFUL, API-Bank, executed BFCL, and executed AppWorld. Appendix F.3 states how the latter two obtain observed results, and appendix C gives their exact provider-free entry points. Before upload, these materials will be supplied as an anonymous archive with hashes and manifests preserved.

REFERENCES

- Sami Abuzakuk, Anne-Marie Kermarrec, Rishi Sharma, Rasmus Moorits Veski, and Martijn de Vos. Optimizing agentic workflows using meta-tools. *arXiv preprint arXiv:2601.22037*, 2026. URL <https://arxiv.org/abs/2601.22037>.
- Lakshya A. Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziemis, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J. Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Alexandros G. Dimakis, Ion Stoica, Dan Klein, Matei Zaharia, and Omar Khattab. GEPA: Reflective prompt evolution can outperform reinforcement learning. In *International Conference on Learning Representations*, 2026. URL <https://arxiv.org/abs/2507.19457>. Oral presentation.
- Anastasios N. Angelopoulos, Stephen Bates, Emmanuel J. Candès, Michael I. Jordan, and Lihua Lei. Learn then test: Calibrating predictive algorithms to achieve risk control. In *International Conference on Learning Representations*, 2022. URL <https://openreview.net/forum?id=TNc0rox3tQ>.
- Vasanth Bala, Evelyn Duesterwald, and Sanjeev Banerjia. Dynamo: A transparent dynamic optimization system. In *Proceedings of the ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, pp. 1–12, 2000. doi: 10.1145/349299.349303.
- Kinjal Basu, Ibrahim Abdelaziz, Kiran Kate, Mayank Agarwal, Maxwell Crouse, Yara Rizk, Kelsey Bradford, Asim Munawar, Sadhana Kumaravel, Saurabh Goyal, Xin Wang, Luis A. Lastras,

- and Pavan Kapanipathi. NESTFUL: A benchmark for evaluating LLMs on nested sequences of API calls. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 33538–33547, 2025. doi: 10.18653/v1/2025.emnlp-main.1702. URL <https://aclanthology.org/2025.emnlp-main.1702/>.
- Yulang Chen, Haoxuan Peng, Jinyan Liu, Zichen Wen, Dongrui Liu, and Linfeng Zhang. AgentSlimming: Towards efficient and cost-aware multi-agent systems. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*, pp. 30064–30086, 2026. doi: 10.18653/v1/2026.acl-long.1387. URL <https://aclanthology.org/2026.acl-long.1387/>.
- James Cheney, Laura Chiticariu, and Wang-Chiew Tan. Provenance in databases: Why, how, and where. *Foundations and Trends in Databases*, 1(4):379–474, 2009. doi: 10.1561/19000000006.
- C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi: 10.1093/biomet/26.4.404.
- Sumit Gulwani. Automating string processing in spreadsheets using input-output examples. In *Proceedings of the 38th ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages*, pp. 317–330, 2011. doi: 10.1145/1926385.1926423.
- Headroom Labs. Headroom: Compressing tool outputs, logs, and retrieved context before they reach the LLM. Software repository, <https://github.com/headroomlabs-ai/headroom>, 2026. Apache-2.0; self-reported 60–95% token reduction on JSON payloads and 73% on a GitHub triage workload. Accessed 7 August 2026.
- Urs Hölzle, Craig Chambers, and David Ungar. Debugging optimized code with dynamic deoptimization. In *Proceedings of the ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, pp. 32–43, 1992. doi: 10.1145/143095.143114.
- Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan, Saiful Haq, Ashutosh Sharma, Thomas T. Joshi, Hanna Moazam, Heather Miller, Matei Zaharia, and Christopher Potts. DSPy: Compiling declarative language model calls into self-improving pipelines. In *International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=sY5N0zy50d>.
- Sehoon Kim, Suhong Moon, Ryan Tabrizi, Nicholas Lee, Michael W. Mahoney, Kurt Keutzer, and Amir Gholami. An LLM compiler for parallel function calling. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 24370–24391, 2024. URL <https://proceedings.mlr.press/v235/kim24y.html>.
- Junyan Li, Zhang-Wei Hong, Maohao Shen, Yang Zhang, and Chuang Gan. FlowCompile: An optimizing compiler for structured LLM workflows. *arXiv preprint arXiv:2605.13647*, 2026. URL <https://arxiv.org/abs/2605.13647>.
- Minghao Li, Yingxiu Zhao, Bowen Yu, Feifan Song, Hangyu Li, Haiyang Yu, Zhoujun Li, Fei Huang, and Yongbin Li. API-bank: A comprehensive benchmark for tool-augmented LLMs. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 3102–3116, 2023. doi: 10.18653/v1/2023.emnlp-main.187. URL <https://aclanthology.org/2023.emnlp-main.187/>.
- John M. Lucassen and David K. Gifford. Polymorphic effect systems. In *Proceedings of the 15th ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages (POPL)*, pp. 47–57, 1988. doi: 10.1145/73560.73564.
- Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M. Waleed Kadous, and Ion Stoica. RouteLLM: Learning to route LLMs with preference data. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/5503a7c69d48a2f86fc00b3dc09de686-Abstract-Conference.html.

- Shishir G. Patil, Huanzhi Mao, Fanjia Yan, Charlie Cheng-Jie Ji, Vishnu Suresh, Ion Stoica, and Joseph E. Gonzalez. The berkeley function calling leaderboard (BFCL): From tool use to agentic evaluation of large language models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 48371–48392, 2025. URL <https://proceedings.mlr.press/v267/patil25a.html>.
- Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, Sihan Zhao, Lauren Hong, Runchu Tian, Ruobing Xie, Jie Zhou, Mark Gerstein, Dahai Li, Zhiyuan Liu, and Maosong Sun. ToolLLM: Facilitating large language models to master 16000+ real-world APIs. *arXiv preprint arXiv:2307.16789*, 2023. URL <https://arxiv.org/abs/2307.16789>.
- Armando Solar-Lezama, Liviu Tancau, Rastislav Bodik, Sanjit Seshia, and Vijay Saraswat. Combinatorial sketching for finite programs. In *Proceedings of the 12th International Conference on Architectural Support for Programming Languages and Operating Systems*, pp. 404–415, 2006. doi: 10.1145/1168857.1168907.
- Harsh Trivedi, Tushar Khot, Mareike Hartmann, Ruskin Manku, Vinty Dong, Edward Li, Shashank Gupta, Ashish Sabharwal, and Niranjan Balasubramanian. AppWorld: A controllable world of apps and people for benchmarking interactive coding agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 16022–16076, 2024. doi: 10.18653/v1/2024.acl-long.850. URL <https://aclanthology.org/2024.acl-long.850/>.
- Zora Zhiruo Wang, Jiayuan Mao, Daniel Fried, and Graham Neubig. Agent workflow memory. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 63897–63911, 2025. URL <https://proceedings.mlr.press/v267/wang25bx.html>.
- Lei Wei, Qi Liu, Ruiyang Huang, Xiao Peng, Sicong Xie, Lanbo Lin, Chenhao Jiang, Yuanwu Xu, Tianyuan Yang, Jiayao Liu, Li Cai, Zhaolu Kang, and Bin Wang. EvoC2F: Compiling tool orchestration for efficient and evolvable LLM agents. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, 2026. URL <https://openreview.net/forum?id=ZSGB91kMOG>.
- Caleb Winston, Ron Yifeng Wang, Azalia Mirhoseini, and Christos Kozyrakis. Agent JIT compilation for latency-optimizing web agent planning and scheduling. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, 2026. URL <https://arxiv.org/abs/2605.21470>.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=WE_vluYUL-X.
- Qizheng Zhang, Michael Wornow, and Kunle Olukotun. Cost-efficient serving of LLM agents via test-time plan caching. *arXiv preprint arXiv:2506.14852*, 2025. URL <https://arxiv.org/abs/2506.14852>.

A PROOF OF PROPOSITION 1

Condition on the train/development data, fix $\eta \in \Lambda$, and write $\gamma = \delta/|\Lambda|$. Let D_i^η and W_i^η be the group admission and violation indicators defined in section 2. Because the entire candidate and grid were frozen before calibration and groups are i.i.d., the pairs (D_i^η, W_i^η) are i.i.d. Conditional on $n_\eta = m > 0$ admitted groups, their violation count therefore satisfies $k_\eta \mid n_\eta = m \sim \text{Binomial}(m, r_\eta)$, where $r_\eta = \Pr(W^\eta = 1 \mid D^\eta = 1)$. Inverting the one-sided exact binomial test (Clopper & Pearson, 1934) gives the upper bound $U_\eta = \text{Beta}^{-1}(1 - \gamma; k_\eta + 1, m - k_\eta)$ of eq. (7), which satisfies $\Pr\{r_\eta > U_\eta \mid n_\eta = m\} \leq \gamma$. When $m = 0$ the event is impossible under the convention that empty-admission thresholds receive the vacuous bound $U_\eta = 1$. Averaging over the random value of n_η therefore yields $\Pr\{r_\eta > U_\eta\} \leq \gamma$ for each fixed threshold. Applying the union bound over the finite grid gives $\Pr\{\exists \eta \in \Lambda : r_\eta > U_\eta\} \leq |\Lambda|\gamma = \delta$, i.e. simultaneous coverage with probability at least $1 - \delta$. On that event, every threshold whose upper bound is at most α satisfies the group-level target in eq. (4), including the threshold returned by Alg. 2. ■

Two scope conditions are worth restating. First, the statement is *per fixed candidate*: Alg. 1 may calibrate several families and retain survivors, and that outer search is not multiplicity-corrected — except when the search is frozen, per the corollary below. Second, i.i.d. source groups is an assumption about the calibration cohort, not a property the compiler verifies; the grouped split in Alg. 1 makes it plausible by construction but does not establish it under temporal drift.

Corollary 2 (Compiler-wide guarantee under frozen selection). *If Alg. 1 runs with candidate freezing enabled — ranking on train data, synthesizing and challenging on train and development windows only, and halting the search the instant any candidate reaches calibration, whatever its outcome — proposition 1’s guarantee holds for the run’s actual reported output with no adjustment to $\gamma = \delta/|\Lambda|$: no candidate multiplicity remains to correct for.*

Proof. Ranking, synthesis, the challenge test, and the frozen score model q are all functions of train and development windows alone, fixed before a single calibration window is inspected. The search reaches calibration for the top-ranked surviving candidate exactly once and halts immediately after that candidate’s outcome, so no second candidate’s program, guard, verifier, or score model is ever compared against calibration data. Proposition 1’s precondition — one candidate fixed before observing calibration — is met by the run’s entire search, not by a hypothetical single-candidate sub-procedure, so its conclusion transfers unchanged. ■ □

This is a corollary, not a new inequality: it is a restatement of proposition 1 for a search that never lets a second candidate see calibration data. It is not hypothetical: the cross-repository extension already runs with freezing enabled, and every sealed repository report there shows exactly one candidate reaching calibration, so corollary 2 applies to those admissions directly. The three primary GitHub families do not use it — each mines two candidates that both reach calibration, with the higher-removal survivor kept by dominance — so they remain the per-candidate certificates above and are not retroactively strengthened. `paper/scripts/frozen_candidate_coverage_simulation.py` backs this with an exact (non-Monte-Carlo) combinatorial check: at $n = 92$, the population rate maximizing a single candidate’s miscoverage is $r^* \approx 0.637$ at miscoverage $\approx 0.00909 \approx \gamma$; pooling m such candidates at the same per-candidate budget with no correction lets the probability that *some* certificate is wrong grow with m (1.8% at $m = 2$ to 13.6% at $m = 16$, crossing $\delta = 0.10$ past the eleven-point grid), while the Bonferroni correction already discussed keeps every tested m under 1%, at the group cost this appendix already reports (106 at $m = 2$, recomputed here independently). A companion seeded Monte Carlo check (seed 20260822, $M=200,000$) draws the 92 groups in correlated blocks instead of i.i.d. and finds realized miscoverage matches the exact baseline at zero correlation but reaches 56% against a 0.9% nominal budget at the most clustered configuration tested, quantifying rather than only asserting why the live studies’ `min_days=min_principals=1` configuration is this guarantee’s weakest link.

B ADDITIONAL ALGORITHMS

Algorithm 3 constructs the typed provenance graph that decides eq. (1). Its two block conditions are asymmetric on purpose: a slot with no witness is treated as a genuine model decision and a slot with

Algorithm 3 BUILDPATG — typed argument provenance. A slot with no witness is a genuine model decision; a slot with too many witnesses is ambiguous. Both block the region rather than defaulting to the cheapest explanation.

Input: episodes $\mathcal{T}$; entry schema Θ ; effect catalog $\mathcal{C}$; transform library $\mathcal{L}$; ambiguity cap κ
Output: provenance graph G retaining $1, \dots, \kappa$ candidate edges per nonliteral grounded slot; declared literals are marked separately

```

1:  $G \leftarrow \emptyset$ 
2: for all episodes  $E = (z, M, e_{1:T}, y) \in \mathcal{T}$  do
3:    $\Sigma \leftarrow \{("z", \pi, v) : (\pi, v) \in \text{FLATTEN}(z), \pi \in \Theta\}$  ▷ only allowlisted entry paths are eligible
4:   for all tool calls  $c_j \in e_{1:T}$  in order do
5:     for all argument slots  $u \in \text{args}(c_j)$  do
6:       if  $u$  is declared literal-only for  $c_j$  then
7:         mark  $(c_j, u)$  LITERAL; continue ▷ preserve the declared constant; infer no provenance
8:       end if
9:        $\mathcal{H} \leftarrow \text{BESTPERSOURCE}\{(\sigma, g) \in \Sigma \times \mathcal{L} : g(\text{value}(\sigma)) = u, \text{depth}(g) \leq 2\}$ 
10:      if  $\mathcal{H} = \emptyset$  then
11:        mark  $(c_j, u)$  MODEL-ORIGINATED; block region
12:      else if  $|\mathcal{H}| > \kappa$  then
13:        mark  $(c_j, u)$  AMBIGUOUS; block region
14:      else
15:         $G \leftarrow G \cup \{\sigma \xrightarrow{g} (c_j, u) : (\sigma, g) \in \mathcal{H}\}$  ▷ synthesis chooses a stable source across groups
16:      end if
17:    end for
18:     $\Sigma \leftarrow \Sigma \cup \{(\text{var}(c_j), \pi, v) : (\pi, v) \in \text{FLATTEN}(o_j), \text{ELIGIBLE}(\pi, v)\}$  ▷ a result witnesses only later
19:  end for
20:  for all pairs  $(e_s, e_t)$  sharing a resource where either writes it do
21:     $G \leftarrow G \cup \{e_s < e_t\}$  ▷ order edge: no reordering is ever proposed across a conflict
22:  end for
23: end for
24: return  $G$ 

```

more than κ witnesses is treated as ambiguous, so the analysis never resolves an argument by picking the cheapest available explanation.

Algorithm 4 is the runtime counterpart of eq. (3). Cheap deterministic checks reject first and the calibrated gate runs only on what survives them, so the common case costs a registry lookup and a guard evaluation. Exactly one exit path compacts; every clean rejection returns the unmodified baseline agent, and any failure whose reversibility cannot be attested raises an incident rather than feigning a rollback. “Baseline” is byte-exact only where a staging owner holds the commit boundary: the outer runner can restore byte-identical model input, whereas a model-boundary adapter cannot un-emit a response the host has already committed to conversation history.

C CONFIGURATION

Live-provider configuration. Every live study in this paper calls one provider and one model: OpenAI gpt-5.6-luna through the openai-agents SDK’s Responses interface. Model settings are fixed across all conditions at reasoning effort low and verbosity low, with parallel tool calls disabled so provider-visible interface counts are comparable between the baseline and a compiled prefix, and server-side response storage disabled. Sampling parameters are not overridden and take provider defaults. Reported dollars are an *estimate*, computed by applying the published standard short-context list rates below — retrieved 2026-08-02 — to the token counts retained per record; they are not an account invoice and carry no negotiated or cached-tier discount beyond the cached-input line.

Algorithm 4 DISPATCH — boundary-time admission. Cheap deterministic checks reject first and the calibrated gate runs only on what survives them. Exactly one exit path compacts; clean rejections return the unmodified agent, while an unattested or post-commit failure raises an incident.

Input: entry state z ; runtime context M' ; registry $\mathcal{R}$; effect catalog C ; mode $m \in \{\text{off}, \text{shadow}, \text{live}\}$
Output: observations for the region, FALLBACK to baseline agent B , or INCIDENT

```

1: if  $m = \text{off}$  then
2:   return FALLBACK                                ▷ conformance: model-visible input must be byte-identical
3: end if
4:  $A \leftarrow \mathcal{R}.\text{RESOLVE}(M'.\text{compatibility}, M'.\text{partition})$ 
5: if  $A = \perp \vee A.\text{lifecycle} \neq \text{ACTIVE} \vee \neg \text{VERIFYSIGNATURE}(A)$  then
6:   return FALLBACK
7: end if
8: if  $H_A(z, M') \neq 1$  then
9:   return FALLBACK                                ▷ manifest pins, isolation keys, typed hulls, effect set
10: end if
11: if  $\text{tools}(A.P) \cap \text{ALREADYOBSERVED} \neq \emptyset$  then
12:   return FALLBACK                                ▷ the region already ran; its live-ins are not the fitted ones
13: end if
14: if  $m = \text{live} \wedge \exists f \in \text{tools}(A.P) : C(f).\text{quota\_attested} \wedge \text{SNAPSHOT} = \perp$  then
15:   return FALLBACK                                ▷ no staging owner, so a discarded attempt is not attestable
16: end if
17: if  $q_A(z) > \eta_A$  then
18:   return FALLBACK                                ▷ calibrated abstention on an uncertain input
19: end if
20: if  $m = \text{shadow}$  then
21:   log would-dispatch; return FALLBACK                ▷ observe coverage without changing behaviour
22: end if
23:  $S \leftarrow \text{STAGE.BEGIN}(); r \leftarrow \text{INTERPRET}(A.P, z, \text{FACADE}(C))$     ▷ the facade re-checks effects at execution
    time
24: if  $r$  raised PreCommitError then
25:   return FALLBACK if  $S.\text{ABORTANDATTEST}()$ , else INCIDENT
26: else if  $r$  raised PostCommitError then
27:   return INCIDENT                                ▷ an external commitment happened; do not feign rollback
28: end if
29: if  $V_A(r) \neq 1$  then
30:   return FALLBACK if  $S.\text{ABORTANDATTEST}()$ , else INCIDENT    ▷ live-outs, effect multiset, and call count
    are checked before use
31: end if
32: return  $r.\text{observations}$  if  $S.\text{COMMIT}(r.\text{effects})$  succeeds, else INCIDENT

```

Rate (USD per million tokens)	gpt-5.6-1una
Input	0.20
Cached input	0.02
Cache write	0.25
Output	1.20

Because every condition within a family shares this model, these settings, the same records, and the same grader, the conditions differ in execution structure alone.

Admission budgets. All reported admission runs use risk budget $\alpha = 0.05$, confidence budget $\delta = 0.1$, and a fixed grid of $|\Lambda| = 11$ thresholds, which fixes the zero-violation sample requirement at $n_\eta \geq \log(\delta/|\Lambda|)/\log(1 - \alpha) = 91.6$, i.e. 92 admitted groups; at exactly $n_\eta = 92$ and $k_\eta = 0$ the bound is $U_\eta = 1 - (0.1/11)^{1/92} = 0.0498 \leq \alpha$. Tightening the confidence budget to $\delta = 0.05$ would raise the requirement to 106 groups, which no study in this paper reaches, so $\delta = 0.1$ is the operative setting throughout and is reported as such rather than rounded to match α . A Bonferroni repair over $m = 2$ candidate families would use $\gamma = \delta/(m|\Lambda|)$ and land on the same 106-group requirement, which is why the paper reports per-candidate certificates rather than a compiler-wide one. The grid is $\Lambda = \{0.02, 0.05, 0.08, 0.11, 0.14, 0.17, 0.20, 0.25, 0.30, 0.40, 0.50\}$ and the ambiguity cap is $\kappa = 3$. Synthesis uses the 23-operator closed library of table 3 at depth ≤ 2 ; region mining uses $w_{\max}$ tool calls per window and requires support across independent scenario groups.

Table 3: The closed transform library $\mathcal{L}$ (version T-v1). These 23 operators are the *entire* expressive power of the binder: a slot with no chain of length ≤ 2 over a single source is ungroundable and retires its region, so this table fixes what eq. (1) can witness. Parameters shown in parentheses are drawn from the observed trace, not free-searched. Every chain is typed, and a runtime type mismatch prunes it during the value-directed search rather than at apply time.

Class	n	Operators
Identity / coercion	5	id, str, int, float, bool
String	6	lower, upper, strip, split(sep)[i], join(sep), fmt(template)
Numeric	5	add(c), mul(c), round(k), sum, len
Collection	6	project(path), filter(path op const), first, last, sort(path), topk(path, n)
Temporal	1	date_fmt(pattern)
Total	23	

The library is closed and versioned deliberately. Every operator added widens Alg. 3’s coverage and its spurious-match surface together, so growth is a two-sided trade rather than a free improvement; artifacts pin the library version and a runtime version mismatch is a dispatch barrier. The search is value-directed rather than an enumeration of $|\mathcal{L}|^2$ expressions: it runs breadth-first from the concrete source value and keeps chains whose denotation equals the target, which prunes by runtime type for free and collapses observationally equivalent chains.

The nonconformity score q . q is an ℓ_2 -regularized logistic model over entry-observable features only — missing-field indicators, empirical-hull margin, temporal drift, and provenance ambiguity — standardized on the development split’s own moments so no calibration statistic enters the fit. Its positive class is an *unproductive* development group: one the artifact does not carry cleanly to a verified result, which on $\mathcal{D}$ means a guard rejection, a verifier abstention, or a manifest mismatch. Wrong executions are not available as training signal at this point in Alg. 1, because `REPLAYANDCHALLENGE` at line 12 retires any candidate that produces one on $\mathcal{D}$; every candidate reaching `FITSORE` at line 13 therefore has zero contract violations on the development split by construction. This is a design consequence rather than an oversight — the cascade is ordered so that deterministic checks reject before the statistical stage runs — but it does mean q is fitted on abstention-shaped supervision and asked at calibration to rank a different event, which is one plausible reason it discriminates weakly (section 7).

What cohort size permits the gate to certify. The zero-violation floor $n_\eta \geq 92$ constrains coverage, not just admission. Certifying selective coverage $c = n_\eta/|\mathcal{K}|$ requires $|\mathcal{K}| \geq \lceil 92/c \rceil$: 92 groups for $c = 1$, 115 for $c = 0.8$, 132 for $c = 0.7$, 184 for $c = 0.5$. Every cohort in this paper fixes $|\mathcal{K}| = 92$, which is the smallest pool at which eq. (7) is satisfiable at all. At that size the admissible coverage set is exactly $\{0, 1\}$: abstaining on one group leaves $n_\eta = 91$ and $U_\eta = 1 - (0.1/11)^{1/91} = 0.0503 > \alpha$. The step-like gate profiles reported in section 5.3 and appendix H are therefore consistent with two different worlds — a score that cannot rank, and a cohort too small to certify a ranking — and these experiments do not separate them. A frontier study must first buy calibration groups. The implementation also supports a minimum-distinct-days requirement, which the single-snapshot live study does not exercise (`min_days = 1`). Two terms of eq. (5) are coarser in the implementation than the notation suggests: ρ_{eff} is approximated by the fraction of tool names containing a namespace separator rather than read from the effect catalog, and $|F|$ is substituted by the argument-slot count of the first observed window because ranking precedes synthesis. Both appear only in a ranking heuristic — no admission decision depends on either, and the effect catalog is consulted where it is load-bearing, in qualification and in the runtime facade.

The two executable-gold external substrates are provider-free after their pinned sources are prepared. Their exact compiler entry points are:

```
B=paper/scripts/bfcl_compiler_benchmark.py
python "$B" --source-root "$BENCHMARK_SOURCE_ROOT"
A=paper/scripts/appworld_compiler_benchmark.py
python "$A" --source-root "$BENCHMARK_SOURCE_ROOT" \
  --appworld-python "$APPWORLD_VENV/bin/python"
```



```
D=paper/scripts/appworld_dispatch_preflight.py
python "$D" --source-root "$BENCHMARK_SOURCE_ROOT"
```

The AppWorld environment uses Python 3.12 with package version 0.1.3.post1 and data version 0.1.0; the signed source manifest and requirements file pin the remaining acquisition inputs. The BFCL source is pinned to revision 6ea57973c7a6. These commands regenerate only aggregate, non-sensitive evidence; no model or provider call is made.

D PAIRED STATISTICS FOR THE PRIMARY FAMILIES

Per-record paired differences (compiled minus baseline) on the two families that admit a deep prefix, with 10,000-sample bootstrap intervals and two-sided Wilcoxon signed-rank tests over $n = 30$ paired held-out records:

Family	Endpoint	Paired difference (95% CI)	p
PR-outcome	Total tokens	-2196.7 [-2211.1, -2182.5]	1.7×10^{-6}
	Wall latency	-3889 ms [-4499, -3346]	1.9×10^{-9}
	Estimated cost	$-\$5.11 \times 10^{-4}$	1.7×10^{-6}
Backlog	Total tokens	-2315.0 [-2366.0, -2224.1]	1.7×10^{-6}
	Wall latency	-4428 ms [-5400, -3558]	1.9×10^{-9}
	Estimated cost	$-\$5.44 \times 10^{-4}$	1.7×10^{-6}

Provider-request differences are deterministic by construction (every pair removes the same number of turns), so no rank test is reported for them: the statistic would measure the degeneracy rather than the effect.

Table 4: The hand-written comparator priced on the same resource axes as table 1, as percentage reduction against the same baseline and the same held-out records. *Manual* is written with full knowledge of the workflow, so this is the pessimistic comparison for the compiler: it ties on provider requests, issues fewer tool interfaces, and is marginally cheaper in tokens and dollars, while compilation leads only on wall latency (bold). The interface gap is concentrated in issue-type routing, where the compiled prefix covers two reads but the agent still issues the third. We therefore claim automated discovery and guarded admission, not runtime dominance over code written by someone who already knows the workflow.

Condition	Requests	Interfaces	Tokens	Latency	Cost
<i>Issue-type routing</i>					
Compiled	50.0	0.0	39.5	51.7	32.0
Manual	50.0	66.7	58.2	47.2	37.5
<i>PR-outcome audit</i>					
Compiled	75.0	66.7	80.8	73.0	75.3
Manual	75.0	66.7	80.8	65.7	75.2
<i>Backlog-attention routing</i>					
Compiled	74.8	66.3	81.4	68.9	75.1
Manual	74.8	66.3	81.2	73.6	74.2
<i>Weighted total ($n = 90$)</i>					
Compiled	66.6	44.2	63.1	64.2	58.7
Manual	66.6	66.5	71.1	62.2	60.4

E ADDITIONAL EXPERIMENTAL DETAIL

Primary workflow families. The three-family result is the main paper’s strongest live evidence because it combines distinct tool vocabularies, exact task graders, balanced held-out cohorts, and provider-backed execution. The issue-type family intentionally retains a shallower two-read prefix; the newer pull-request and backlog families admit deeper pre-model compaction. This difference

Table 5: Primary GitHub benchmark at a glance. Each sample supplies one record number from a pinned public snapshot; every listed answer field is checked against the source record.

Task	Read-only evidence path	Exact answer contract
Issue type	<code>issue_number</code> → record, official labels, comments	Title, state, label-derived category, verbatim excerpt
PR outcome	<code>pr_number</code> → PR record, merge status, discussion	open, merged, or closed_unmerged, plus title and comment excerpt
Backlog attention	<code>issue_number</code> → open-issue record, assignees, discussion	owned, discussed_unowned, or awaiting_first_response, plus title, owner, and excerpt

explains the lower resource savings on the first family and is part of the evidence boundary, not noise to average away.

Cross-repository extension. The cross-repository study is narrower than the primary workflow families. Its task uses two read-only tools and a simpler exact outcome contract, which is precisely why a fixed template can remain tied or even more efficient than the learned artifact. We keep the result in the main text because it addresses a reviewer-critical scope question—whether guarded specialization survives beyond one repository—while being explicit about the reduced task complexity.

The position invariant, empirically. An early pilot dispatched a compiled *suffix* region at the entry boundary, violating eq. (2). Because the runtime resolves artifacts before the agent has made the calls the region assumed had already happened, the pilot reordered and duplicated tool calls. The failure is a semantic mismatch between compilation-time position and resolution-time position, not a tuning error, which is why eq. (2) is a hard constraint in the deployed compiler rather than a scoring penalty.

Omitted from the main claim. The repository retains additional studies that are useful scientifically but less suitable for the main narrative:

- a natural-order issue study that exposed a factual continuation miss for a more aggressive three-read artifact,
- exploratory guarded composite synthesis and a follow-up comparison to a fair pre-model manual program,
- a bounded GEPA prompt-optimization study in which GEPA retained its seed, so the combined arm is a replication rather than evidence of interaction, and
- a portfolio pilot and controlled demonstration suite for runtime boundary coverage, including workloads where removing turns *increases* cost through prefix-cache fragmentation.

These studies remain important to the broader artifact, but they either use a weaker risk level, address a narrower engineering question, or are explicitly exploratory.

F EVIDENCE BREADTH AND BOUNDARY AUDITS

The main text uses the nine-page budget for the registered live studies, the time-forward extension, and the four compiler-measured external substrates. This appendix makes the retained breadth visible without pooling incomparable cohorts or promoting screened and exploratory work to deployment evidence.

F.1 REPOSITORY-LEVEL TIME-FORWARD OUTCOMES

Why pytorch/pytorch retires in the core protocol and admits in the balanced rerun. The core retirement is attributable to one stage. Its compile report records 116 windows accepted, no windows blocked and none blocked by tool, one candidate mined, and a single rejection at `calibrate:retire` — stage 6 — with the message that the tightest attainable Clopper–Pearson upper bound over the grid is $1.000 > \alpha$. A bound of exactly 1 is eq. (7)’s degenerate branch, reached

Table 6: Disaggregated cross-repository time-forward outcomes behind section 5.2. The core protocol uses four completed repositories and one compile-time retirement; the balanced rerun changes the selected repositories and class mix, so it is reported separately rather than pooled with the core protocol. Reductions compare the compiled condition with the paired unchanged baseline.

Protocol	Repository	Held-out	Exact	Req. ↓	Tokens ↓
Core	huggingface/datasets	30	30/30	44.4%	52.2%
Core	pandas-dev/pandas	30	30/30	44.4%	52.5%
Core	psf/requests	30	30/30	44.4%	52.3%
Core	streamlit/streamlit	30	30/30	44.4%	52.7%
Core	pytorch/pytorch	—	RETIRE	—	—
Balanced	pandas-dev/pandas	60	60/60	66.7%	78.6%
Balanced	psf/requests	60	60/60	66.7%	78.7%
Balanced	pytorch/pytorch	60	60/60	66.7%	78.0%

when no calibration group is admitted at a threshold or when every admitted group violates, so the candidate never reached a countable zero-violation cohort at any η . Provenance, synthesis, and contract induction are therefore not implicated: the region was groundable and the program was synthesized. What differs between the protocols is the class composition of the pool the artifact was fitted and calibrated on. The core protocol draws discovery from the older eligible pool with class balance explicitly not enforced, which for this repository yields 84 `closed_unmerged`, 30 `open`, and 2 `merged` records out of 116, while the sealed held-out split is an exact round-robin of 10/10/10. The deployment distribution the gate is asked to certify therefore contains a third `merged` records against a discovery pool containing under 2% of them. The balanced rerun enforces the `open/merged/closed_unmerged` mix at selection and the same task admits. We report both rather than only the rerun: the core outcome is the protocol behaving as designed under a skewed pool, and it is the clearest instance in the paper of the gate refusing on evidence composition rather than on trace structure.

The core result succeeds on four repositories (120/120 held-out pairs) and retains the fifth as a compile-time retirement after 116 exact discovery traces. The balanced rerun has a distinct three-repository cohort and reaches 180/180 held-out pairs. These rows show breadth across repositories, not independent evidence for an average effect: all runs share the same provider family and the same narrow two-read outcome task.

F.2 CONTINUATION AUDIT

The earlier natural-order three-read study is excluded from the registered $\alpha = 0.05$ headline because it used $\alpha = 0.10$ and exposed a downstream continuation error. Its retained counterfactual replay nevertheless tests a boundary the exact tool contract cannot see: 17/18 candidate continuations pass unchanged, while issue 6602 has a comment-evidence mismatch. Checked rendering repairs that one case, yielding 18/18 final contracts. This audit makes no provider calls and does not estimate live agent quality; it establishes only that the continuation check caught a concrete failure the compacted tool trace alone would miss.

F.3 EXTERNAL EVIDENCE MAP

Table 8 records every named external benchmark in the retained evidence register and its admissible interpretation.

Obtaining observed results from BFCL. BFCL’s reference plans carry no tool results, which is why the plan-format row and the compiler row are listed separately. Its multi-turn categories, however, ship an executable stateful backend: each task publishes an `initial_config` snapshot and the classes it involves, so executing the *official gold plan* through the upstream helper yields observed results, an entry snapshot, and an in-process replay oracle, with no model in the loop. Five preconditions fail the run closed rather than publishing: an undeclared method, any upstream execution error, a read-like declaration observed mutating state, a compilable declaration observed advancing the scenario RNG, and any inexact re-execution. All held: 200 tasks and 1,142 calls executed cleanly

Table 7: Structural dispatch eligibility of the admitted AppWorld artifact on the 28 released official baseline runs. Eligibility is the admitted program’s exact call sequence at normalized position 0; it is an upper bound on φ , because the manifest check and the calibrated gate are not evaluated and the released logs retain API calls rather than model boundaries. Trajectories repeat tasks across runs and are descriptive rather than independent samples. A miss abstains to the unchanged agent.

Released baseline architecture	Runs	Trajectories	Eligible at pos. 0	Rate
Full code + reflection	8	2,340	2,339	1.000
Iterative parallel function calling	4	1,170	762	0.651
Plan and execute	8	2,340	1	0.000
ReAct	8	2,340	0	0.000
Pooled	28	8,190	3,102	0.379

and a second independent pass reproduced every result byte for byte, executing turn-at-a-time where the first pass executed call-at-a-time.

Why the catalog is signed rather than inferred. The same run audits the declarations empirically by snapshotting every involved instance around every call. It observed 1,060 state-mutating calls over 44 of the 81 methods and 94 calls that advanced the seeded scenario RNG, with zero cases of a read-like declaration mutating state or a compilable declaration advancing the RNG. One case is worth naming: a method whose name and docstring present it as a cost getter mutated its instance’s internal lookup on all 36 of its calls and is therefore declared a write. A name- or docstring-derived effect label would have admitted a write into a compiled region. Two further methods are read-like but carry no capabilities, because one derives an age from the host date and the other draws from — and advances — the seeded RNG; they are barriers by declaration and account for 35 suppressed spans. Write barriers suppress 2,802 spans in total, against 45 for entry-schema admissibility, 31 for slot ambiguity, and 5 for partial runs.

Obtaining observed results from AppWorld, and why it is the reachable case. AppWorld’s shipped `api_calls.json` records method, url, and request data but no responses, so it fails closed for the same reason BFCL’s plans do. Executing the official `compiled_solution_code` against the pinned `0.1.3.post1` backend supplies the results; each task’s own `specs.json` supplies the entry snapshot; a second execution supplies the replay oracle. Calls are captured at the single dispatch point every `apis.<app>.<api>(...)` call passes through, so the recorded identity is a semantic app/API pair with typed keyword arguments and a decoded JSON result. Of the 147 public train+dev gold solutions, 136 execute cleanly; 11 dev solutions raise against the shipped data version and are reported as upstream compatibility outcomes rather than dropped. The second pass reproduced 7,070/7,070 results byte for byte, minted access tokens included, because AppWorld freezes each task’s clock.

The empirical audit attributes every INSERT/UPDATE/DELETE/ CREATE/DROP statement to the enclosing call. It observed 1,320 mutating calls, zero read-like declarations mutating, and—unlike BFCL—zero APIs that mutate on some calls but not others. Every GET was clean on every call and every DELETE and PATCH mutated on every call, so the effect boundary is crisp; the five `*.login` endpoints are the only non-GET APIs that never mutate, which is why they are the single contestable declaration and why the run is pre-registered in two arms. Groups are tasks, not scenarios: AppWorld’s three variants of a scenario share a supervisor and a database, so the bound is conditional on task-level independence, which this corpus does not establish—the same conditional the live studies carry, neither weakened nor repaired here. The calibration share is fixed at exactly the 92 groups the bound needs, since a larger set would make the bound easier and a smaller one would make it unreachable. Two held-out replay figures appear for AppWorld and they count different partitions, which table 2 flags but does not have room to explain. The 34/0/0 in that table is the test side of the *family* partition used to challenge a mined candidate — 102 train and 34 test groups, the same accounting every substrate row reports. The 11/11 quoted in section 5.3 is the sealed test split of the 11/92/11 dev/calibration/test partition Alg. 2 consumes for admission. Both are genuinely held out; they are held out from different fits, and neither is a subset of the other’s training side.

Table 8: Disposition of every retained external benchmark family. Only the four compiler-measurement rows support compiler claims; the remaining rows document why broader benchmark names are not silently converted into GAC results. BFCL appears twice because its reference plans and its executed gold plans are different evidence.

Benchmark	Evidence tier	Retained result / claim boundary
NESTFUL	Compiler measurement	1,415 traces; 24 / 12 / 0 replay; all families retire.
API-Bank	Compiler measurement	212 traces; 0 / 2 / 0 replay; all candidates retire.
BFCL v4 (plans)	Gold-plan checker	200/200 valid plans; plan format only.
BFCL v4 (executed)	Compiler measurement	200 traces, 1,142/1,142 exact re-execution; 3 / 3 / 0 replay; all families retire.
AppWorld (executed)	Compiler measurement	136/147 traces, 7,070/7,070 exact re-execution; $n_{\max} = 136 \geq 92$; one artifact admitted at $U = 0.0498$; four candidates retire at the margin.
ToolSandbox	Live-provider subset	One simulated scenario; compiler not run.
τ^2/τ^3	Live-provider subset	0/4 pass; compiler not run.
ToolBench	Adapter smoke	Ten fixtures; full data/backend unavailable.
AgentBench	Adapter/preflight	Services and Freebase data gated.
GAIA	Access preflight	Authorization denied; no metric.
SWE-bench Verified	Task adapter	Official run host-gated.
BrowseComp	Live-web subset	1/3 correct, 28 searches; bypassed by the compiler.

Admissible is not dispatchable. Table 7 separates the two. The admitted region is eligible on essentially every full-code trajectory and essentially no ReAct or plan-and-execute trajectory, because those architectures open by reading API documentation—3,752 of the 8,190 released trajectories begin with an API-description call—so the compiled prefix is no longer at position 0 and (2) refuses. A further 61 trajectories contain the program somewhere other than position 0, which the same invariant refuses. Full-code agents are eligible on 2,339/2,340 trajectories and iterative parallel function-calling agents on 762/1,170. Two secondary observations: the rate is flat across task difficulty (0.377 on `test_normal` against 0.380 on `test_challenge`), so this is a calling-discipline property rather than a workload property; and the maximal compilable depth is 2 on the eligible families, so the admitted two-step program sits at the ceiling the effect catalog permits rather than short of it. On the ineligible families the measured depth is a floor, not an estimate, because the API-documentation call is undeclared in the signed catalog and unknown effects are not reads.

F.4 WHAT THE REGISTERED GATE HAS AND HAS NOT DEMONSTRATED

This audit strengthens the evidentiary record precisely by preserving the negative result: the registered gate currently behaves as a support threshold, not as a demonstrated risk-coverage frontier. The exploratory partial frontiers therefore remain diagnostic rather than evidence for the main claim.

G EXTENDED LIMITATIONS

Repairs for non-exchangeable groups. Section 7 states that the cluster-aware repairs all cost effective sample size. Concretely, three are compatible with eq. (7) and none is exercised here: enforcing temporal separation at calibration (`min_days > k`) so a burst of related records cannot populate one split; replacing the per-group binomial count with a cluster-level or block-bootstrap bound that treats a correlated burst as a single unit; and carrying the cluster structure into the conformal step directly. Each shrinks the number of independent units below the 92 that the bound already demands, which is why none was affordable on these cohorts and why a larger corpus is the precondition for all three.

Relaxing the position invariant for a read-only prologue. Section 5.3 shows eq. (2) costs essentially all ReAct and plan-and-execute eligibility, because those architectures open with an API-documentation read. Admitting a bounded read-only *prologue* before the compiled prefix looks tractable and is narrower than general suffix dispatch: the adapter would need to recognize the prologue’s calls as already issued and suppress their replay, rather than reconcile an arbitrary mid-trace entry point with whatever the agent has already done — which is the fault appendix E records. The

Table 9: Gate-behaviour inventory from the retained threshold analysis. A partial frontier would show graded coverage as the threshold changes; none appears among the seven current registered artifacts.

Evidence set	Artifacts	Partial frontiers	What it licenses
Current registered $\alpha = 0.05$	7	0	Exact per-candidate certificates; all are step-like.
Exploratory or looser- α	3	2	Diagnostic evidence only; not the registered claim.
NESTFUL refusal	12 synthesized	0	$n_{\max} = 26 < 92$; no admission.
API-Bank refusal	2 synthesized	0	$n_{\max} = 8 < 92$; no admission.
BFCL v4 refusal	4 synthesized	0	$n_{\max} = 15 < 92$; no admission, pre-registered.
AppWorld admission	1 admitted / 4 retired	0	$n_{\max} = 136 \geq 92$; the one reachable external bound. Admitted gate is step-like; retirements are marginal ($U = 0.051$ at 89 and 90 eligible groups).

obligation is replay suppression under an extended entry contract, not a new statistical argument, since the compiled region itself would be unchanged. We have neither built nor tested it, so it is a conjecture with a named mechanism.

Why the workflow-reuse line is not a drop-in comparator. Section 7 states that we run no head-to-head against automated workflow reuse, plan caching, or call scheduling. The obstruction is that each optimizes a different quantity than the one our metric and our gate are defined over. Wang et al. (2025) induces reusable workflows into the *prompt*: the agent still reaches a model decision at every tool boundary, so the provider requests we count are not removed and there is no substituted region to guard. Kim et al. (2024) reschedules the calls of a single episode into a parallel DAG, which lowers latency without eliminating a model turn, and its planner remains in the loop. Zhang et al. (2025) reuses plans across episodes, which is the closest target, but without argument-level provenance or an effect declaration it makes no admissibility decision — so there is nothing to compare our gate against, only a cache-hit rate. Re-expressing any of them as a guarded pre-model substitution with an entry contract and a verifier would change the method being measured rather than benchmark it. The comparison we owe and did not run is instead internal: an *unguarded* arm that replays the mined region deterministically with provenance, effect, and position barriers disabled, which would price the guards rather than the compilation. It is the single highest-value addition to this evaluation.

A compression comparator ran, and did not engage. Every headline resource number compares a compiled prefix against an *uncompressed* baseline, because deployed compression middleware reports token reductions of the same order on a comparable workload by an entirely different mechanism (Headroom Labs, 2026). A live Headroom v0.5.18 ablation adds `headroom_only/compiled_headroom` arms to both GitHub families on the existing 30-record held-out cohort (paper/supplementary/headroom-ablation-protocol.md). Headroom attempted every eligible JSON tool result and pre-model evidence object and applied *none* of them, saving zero tokens in either family, with all four paired quality contrasts holding 30/30 at exact $p=1$. This is a null engagement at this payload shape, not a measured composition or a measured floor — it does not establish Headroom’s behavior on larger contexts — but it does retire the narrower concern that our reduction is compression under another name: on this workload the two systems visibly do not touch the same bytes.

A drift-parity protocol, pre-registered and not yet run. The strongest standing objection is that a hand-written program reaches the same 90/90 contracts as the compiled artifact and is cheaper on one family. A three-arm design isolates the *induced verifier* from the *program* by pairing the compiled artifact with a permissive no-op verifier and testing both against the hand-written program under the nine-family metamorphic perturbation suite already implemented in `evaluation/perturb.py` (reordering, duplication, formatting, empty/null fields, schema drift, tool errors and timeouts), asking whether the compiled artifact abstains where an unverified program would answer silently

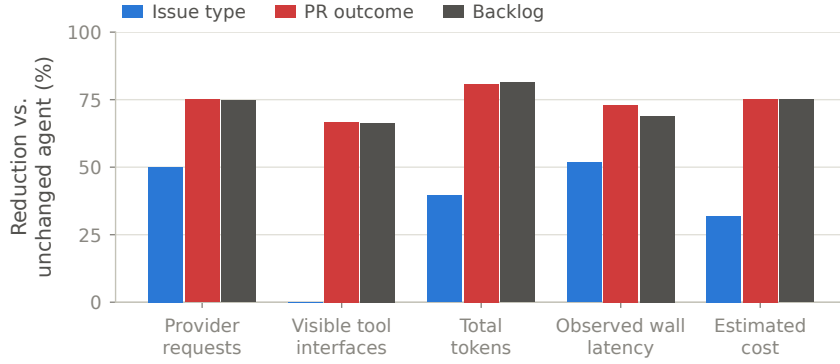


Figure 4: Per-family resource reductions behind table 1, plotted because the spread is the result: admissible prefix depth — not the optimizer — sets the savings, and the two deeper families sit at the feasibility ceiling of eq. (6).

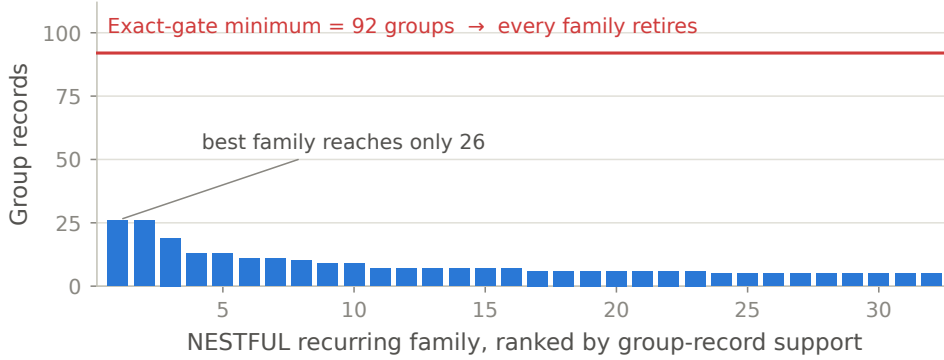


Figure 5: NESTFUL family supports against the 92-group requirement implied by eq. (7) at $\alpha = 0.05$, $\delta = 0.1$, $|\Lambda| = 11$. The largest family reaches 26 groups, so every family retires at calibration even though provenance, synthesis, and replay all succeeded.

(paper/supplementary/drift-robustness-ablation-protocol.md). Status: pre-registered on the deterministic demonstration substrate, not executed; no result from it is reported here.

Where “unchanged baseline” is exact. Pre-emission rejection is exact on both runtime paths; only a staging-owning runner can restore byte-identical model input after an attempt has been emitted (appendix B). Without one, a post-commit verifier failure is an incident rather than a fallback, which is why the compiled region is confined to the pre-commit prefix.

How the calibration cohort must be sampled. The calibration cohort must be sampled before filtering on whether the baseline happened to exhibit the mined region, with violation labels defined for every dispatch-eligible member; post-selecting only recurrent successes would not satisfy proposition 1. This is why the AppWorld substrate fixes its calibration share at exactly the 92 groups the bound requires and draws them by seeded shuffle over all clean tasks, rather than taking the tasks in which the mined family happens to have the deepest support.

Figure 4 plots the per-family spread behind table 1: admissible prefix depth, not the optimizer, sets the savings. The current empirical gate does not yet demonstrate a useful risk–coverage frontier at the registered 5% level. The strongest current runs are mostly all-or-none support tests (fig. 5), and the compiler-wide candidate-search problem is multiplicity-corrected only when the search is frozen (corollary 2); the primary families’ dominance search over two candidates is not. Likewise, exact task contracts do not certify full semantic equivalence of the final answer, especially once a downstream model continuation is involved. Two further operational caveats belong in the record: the headline live artifact was compiled without a sandbox available to the perturbation suite, so it

records `perturbations_claimed: false` rather than a completed challenge, and registry signature verification was configured off in the reported run. These are not hidden defects in the repository; they are the reasons the paper avoids stronger deployment claims.

H PROSPECTIVE GATE-FRONTIER PROTOCOL

`paper/supplementary/prospective-gate-frontier-protocol.md` pre-registers the experiment that could turn section 7’s step-like gate into a genuine risk–coverage frontier. **Status: PRE-REGISTERED, THEN PARTIALLY EXECUTED.** The design below was committed before any repository or provider had been acquired under it, so it could not be revised in light of results it did not yet have; the cohort has since been sealed and executed on four of five repositories, reported at the end of this section without changing anything committed here.

Cohort. At least three repositories or domains, extending the provider-free `github_multirepo_preflight.py` harness, with at least 300 sealed held-out paired cases (zero compiled-only failures in 300 gives a one-sided 95% upper bound near 1%, well below the current small-cohort bounds), reported pooled and per-repository. **Compilation** freezes candidate selection throughout, so any admitted gate inherits corollary 2 rather than a per-candidate certificate. **Three arms** compared under identical records, model, cache policy, and ordering: the unchanged agent, the learned gate of section 2, and a support-only frequency gate. **At least three distinct nonzero coverage levels** must be observed for the learned gate, which requires a cohort deliberately including record classes expected to sit at different points on a certification-difficulty gradient, not two endpoints from a homogeneous task. Thresholds, features, splits, hypotheses, and the stopping rule are fixed and hashed before acquisition.

Decision rule. A frontier with wrong-dispatch at or below the registered bound at every level, dominating the support-only gate at matched coverage, is the predicted positive outcome. A gate that stays step-like while the support-only comparator does not (or vice versa) is a reported negative for the learned score specifically. Any held-out wrong dispatch on either gate is an adverse finding reported in full, ahead of every other reading. Two extensions are named and explicitly deferred pending their own feasibility bar: model/provider breadth, and an executable workflow-compiler comparator (AWO is the closest published candidate), run only if it can match records, model, execution placement, overhead accounting, cache policy, and grading without hand-waving — a forced comparator would weaken the paper more than omitting one.

A pilot for this protocol has since run (`gate-frontier-pilot-protocol.md`, in the same directory): 90 records stratified by whether an issue’s evidence comments contain a Markdown-style link, the one mechanism this codebase has recorded for a compiled artifact’s excerpt to diverge from its source. Violations concentrated there (4/60 against 0/60 at a matched baseline, one-sided $p = 0.059$) while a bare nearby URL predicted nothing (1/60, $p = 0.50$); one issue failed under both the unchanged agent and the compiled artifact independently. A weak go signal, not a frontier: it refines the calibration feature for the protocol above rather than substituting for it.

The comparator feasibility bar has also been checked (spike memo in the same directory): AWO (Abuzakuk et al., 2026) has no confirmed public implementation to build against, so this work-stream closes on that no-go rather than on a forced, unverifiable reimplement.

H.1 OBSERVED RESULTS

The sealed cohort targets 300 pooled held-out pairs across the five repositories section 5.2 already uses, 116 discovery and 60 held-out cases each under frozen-candidate selection. Discovery reaches 580/580 exact traces. Four repositories admit an artifact and complete 240 held-out triples under all three arms; `pytorch/pytorch` again retires at compile time — the same repository, under an independent five-times-larger cohort, reproducing rather than revealing an artifact of the smaller core cohort’s retirement. The pooled result is therefore 240 of the pre-registered 300 pairs, short of target for the stated reason, not concealed.

On the 240 pairs, baseline, learned-gate, and support-only conditions all pass exact task contracts (240/240, 239/239 after one held-out record failed under a transient provider timeout independent of condition or repository). Relative to baseline, the learned gate reduces provider requests 44.4%, total

Table 10: Disaggregated outcomes for the gate-frontier study. “Nonzero coverage in sweep” lists every coverage value the frozen Λ grid produces for the learned gate before threshold selection; “Admitted” is the value actually deployed. Support-only ($\alpha = 1$) produces the identical sweep and admission on every repository.

Repository	Held-out	Exact	Nonzero coverage in sweep	Admitted	Req. $\downarrow$
huggingface/datasets	60	60/60	0.1087 (upper 0.375, rejected), 1.0	1.0	44.4%
pandas-dev/pandas	60	60/60	1.0	1.0	44.4%
psf/requests	60	59/60	1.0	1.0	44.4%
streamlit/streamlit	60	60/60	1.0	1.0	44.4%
pytorch/pytorch	—	RETIRE	none (upper 1.0 at every η)	—	—

tokens 52.3%, observed wall latency 51.0%, and estimated cost 48.3%; the support-only gate — $\alpha = 1$ in an otherwise byte-for-byte identical second `compile_gcc` pass, so mining, synthesis, challenge, and freezing are unchanged and only the risk budget is removed — is statistically indistinguishable: 44.4%, 52.4%, 47.9%, and 48.8%. Table 10 shows why: three of four admitting repositories never accept a calibration group below $\eta = 0.11$ and accept all 92 at $\eta \geq .11$, a pure two-point step; the fourth, `huggingface/datasets`, additionally accepts 10 of 92 groups at $\eta \leq 0.08$, but that point’s exact upper bound (0.375) exceeds the registered $\alpha = 0.05$ budget and is never admissible, so every repository deploys the same coverage-1.0 threshold. No repository, under either gate, ever deploys at an intermediate admissible coverage.

Per the decision rule fixed in advance (section 7), this is the fourth listed outcome: neither gate produces graded coverage on this cohort, so the null is reported rather than a manufactured frontier. At four times the prior cross-repository scale, and against a comparator that shares every statistical mechanism except the risk budget, the exact- $\alpha = 0.05$ gate remains a support threshold. This confirms, and does not resolve, the step-gate finding section 7 already reports.